📄 Task-1 Report

Title: Classification Model

1. Objective
Build a classification model to predict cancer diagnosis.

2. Steps Performed

- Loaded dataset
- Checked missing values
- Split data into train/test
- Applied Logistic Regression
- Applied Random Forest
- Performed cross-validation
- Evaluated metrics

3. Models Used

- Logistic Regression
- Random Forest

4. Metrics Used

- Accuracy
- Precision
- Recall
- F1 Score
- ROC-AUC

5. Results

```
mean radius                0
mean texture               0
mean perimeter             0
mean area                  0
mean smoothness            0
mean compactness           0
mean concavity             0
mean concave points        0
mean symmetry              0
mean fractal dimension     0
radius error               0
texture error              0
perimeter error            0
area error                 0
smoothness error           0
compactness error          0
concavity error            0
concave points error       0
symmetry error             0
fractal dimension error    0
worst radius               0
worst texture              0
worst perimeter            0
worst area                 0
worst smoothness           0
worst compactness          0
worst concavity            0
worst concave points       0
worst symmetry             0
worst fractal dimension    0
target                     0
dtype: int64
--- Logistic Regression ---
Accuracy: 0.956140350877193
Precision: 0.9487179487179487
Recall: 0.9866666666666667
F1 Score: 0.9673202614379085
ROC-AUC: 0.9420512820512821

--- Random Forest ---
Accuracy: 0.9736842105263158
Precision: 0.9615384615384616
Recall: 1.0
F1 Score: 0.9803921568627451
ROC-AUC: 0.9615384615384616

LR Cross-Val Accuracy: 0.9507995652848935
RF Cross-Val Accuracy: 0.9578171091445427
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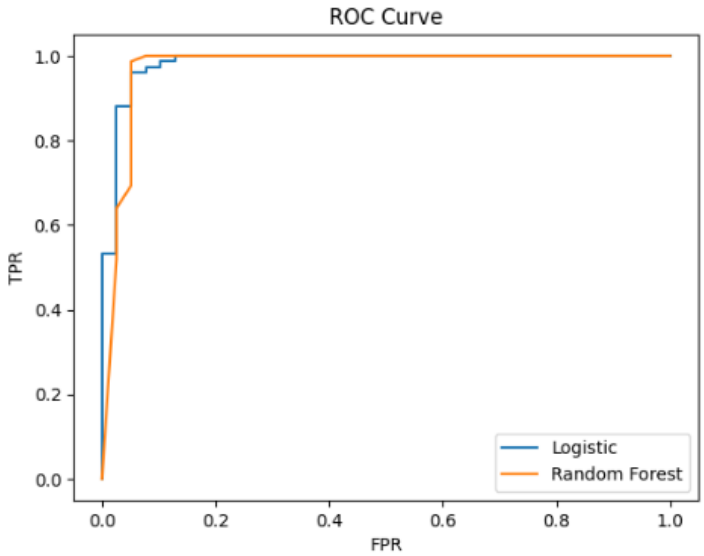

6. Conclusion

(Random Forest / Logistic is better based on output)